

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11.	(a)			profile with start and end at same energies and activation energy of 58 kJ mol ⁻¹ allow tolerance ± 1 small square each way for peak maximum		1		1		
	(b)	(i)		$k = A e^{-E_a/RT}$ accept $\ln k = \ln A - \frac{E_a}{RT}$	1			1	1	
		(ii)		$e^{-E_a/RT} = 1.19 \times 10^{-4}$ (1) ecf possible from part (i) $E_a = 22.5 \text{ kJ mol}^{-1}$ (1) activation energy is lower so this catalyst is more effective (1) ecf from activation energy calculated			3	3	2	
	(c)			$\Delta_f H^\ominus = 2 \times \Delta_f H^\ominus(\text{H}_2\text{O}) - 2 \times \Delta_f H^\ominus(\text{H}_2\text{O}_2)$ (1) $2\Delta_f H^\ominus(\text{H}_2\text{O}_2) = (2 \times -286) - (-98) = -474$ $\Delta_f H^\ominus(\text{H}_2\text{O}_2) = -237$ (1)		1 1		2	1	